STRUCTURE MAPS OF A GROUPOID OVER A FIELD OF CHARACTERISTIC ZERO ARE SMOOTH

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Allowed standard prerequisites (editorial).

- The sheaf of differentials of R to U for a groupoid over a field with s, t locally of finite type is free (more-groupoids chapter, Lemma lemma-sheaf-differentials, canonical tag 04LA; collected on this lane as L22526). cited at proof lines 1631-1633. Provides the freeness of the sheaf of differentials of R to U, the input to the smoothness criterion.
- Over a field of characteristic zero a locally finite type morphism with free sheaf of differentials is smooth (Varieties chapter, Lemma varieties-lemma-char-zero-differentials-free-smooth, canonical tag 04QN). cited at proof lines 1634-1635. Turns freeness of the differentials plus local finite type into smoothness of s and t.

Source reference. Original lemma 04R9: sources/stacks-chow/tags.txt, source lines 6177–6177. Cited at proof lines 1631-1633: the sheaf of differentials of R to U is free.

Source reference. Original lemma 04QN: sources/stacks-chow/tags.txt, source lines 6156–6156. Cited at proof lines 1634-1635: over a field of characteristic zero, free differentials plus locally finite type give smoothness.

Lemma 0.2. *Let S be a scheme. Let (U, R, s, t, c) be a groupoid scheme over S. Assume*

(1) *$U = \mathrm{Spec}(k)$ with k a field,*
(2) *s, t are locally of finite type, and*
(3) *the characteristic of k is zero.*

Then $s, t : R \to U$ are smooth.

Proof. By Lemma 04R9 the sheaf of differentials of $R \to U$ is free. Hence smoothness follows from Varieties, Lemma 04QN. □

Date: stacks-project:more-groupoids.tex:local-snapshot-20261012; GNU FDL 1.2 or later.